



FRAUDSHIELD AI

"Detect. Investigate. Explain. Act."

Agentic AI for Cyber-Fraud Detection, Investigation & Explainable Risk Assessment



USER

Reports a concern



SUSPICIOUS INPUT

SMS · Email · URL · Transaction



AI FRAUD AGENT

Investigates the case



RISK DETECTION

Explainable risk score



PROTECTION

Safe action recommended

PROBLEM STATEMENT ID

[PROBLEM STATEMENT ID]

PS CATEGORY

Software

PROBLEM STATEMENT TITLE

[PROBLEM STATEMENT TITLE]

TEAM ID

[TEAM ID]

THEME

[THEME]

TEAM NAME

[TEAM NAME]

THE PROBLEM

- Scam messages & phishing links are hard for users to verify.
- Existing tools detect but rarely explain evidence or next steps.
- Fraud signals are often checked in isolation, not together.
- Users need technical risk turned into clear, actionable guidance.

OUR SOLUTION

FraudShield AI is an Agentic AI fraud-investigation platform that:

- Accepts SMS, email, URL, phone & transaction inputs
- Classifies the case automatically
- Dynamically selects relevant investigation tools
- Collects structured evidence
- Fuses ML, rules & threat intelligence
- Generates an explainable risk assessment
- Recommends safe next actions

HOW THE AGENT INVESTIGATES A CASE



Core innovation: the agent performs Dynamic Tool Selection instead of running a fixed pipeline.

INNOVATION & UNIQUENESS



Agentic Investigation

AI dynamically selects tools based on the case.



Evidence Fusion

Combines ML, rules and threat intelligence.



Explainable Risk

Shows why a case was flagged.



Human-in-the-Loop

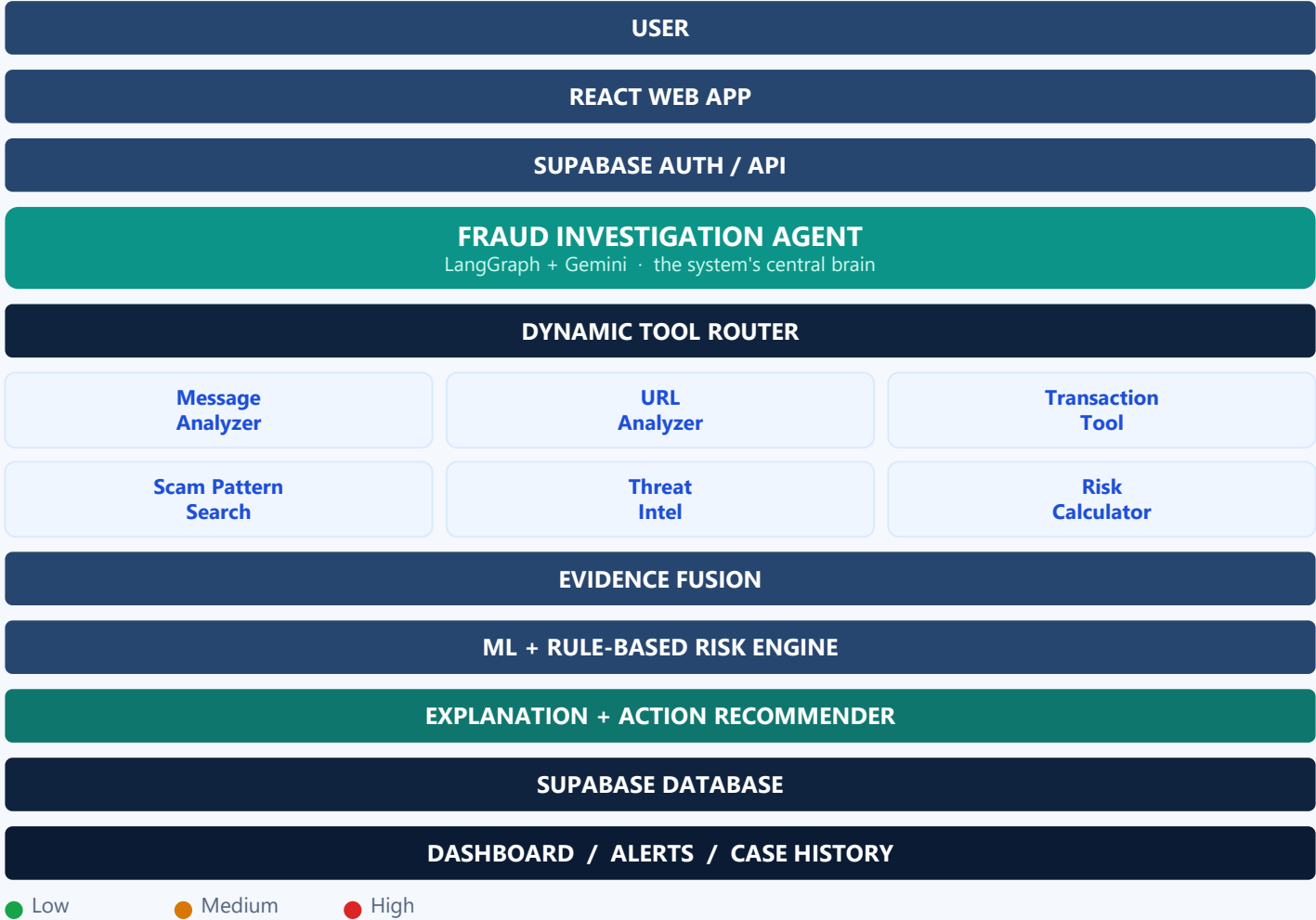
High-risk cases reviewed before action.



Continuous Feedback

False-positive feedback improves the system.

SYSTEM ARCHITECTURE



TECH STACK

FRONTEND	React · TypeScript · Tailwind CSS · shadcn/ui · Recharts
BACKEND	Supabase · PostgreSQL · Edge Functions
AI	Gemini API · LangGraph
ML	Python · Scikit-learn · Pandas · NumPy
DATA	UCI SMS Spam · UCI Phishing URLs · Credit Card Fraud (Kaggle/ULB)
THREAT INTEL	PhishTank (where available)
DEV	GitHub · VS Code · Claude Code (AI-assisted)
DEPLOY	Cloudflare Pages · Supabase

IMPLEMENTATION FLOW

- 1 Input received
- 2 Input classified
- 3 Relevant tools selected
- 4 Evidence collected
- 5 Risk calculated
- 6 Explanation generated
- 7 Action recommended
- 8 Case stored

TECHNICAL FEASIBILITY

- Mature, open-source technologies throughout
- React + Supabase simplify application development
- LangGraph supports stateful agent workflows
- Modular tool architecture — new analyzers plug in easily

OPERATIONAL FEASIBILITY

- Simple web interface for non-technical users
- Investigation results explained in plain language
- High-risk cases flagged for manual review
- Cloud architecture supports centralized case management

ECONOMIC / SCALABILITY VIABILITY

- Prototype buildable primarily with free / open-source tools
- Serverless architecture reduces infrastructure complexity
- Modular services allow independent scaling

CHALLENGES & MITIGATION

False positives

Human review + feedback loop + threshold tuning

Limited threat-intelligence availability

Local indicators + configurable external adapters

AI hallucination

LLM explains structured evidence — does not invent it

API / rate limits

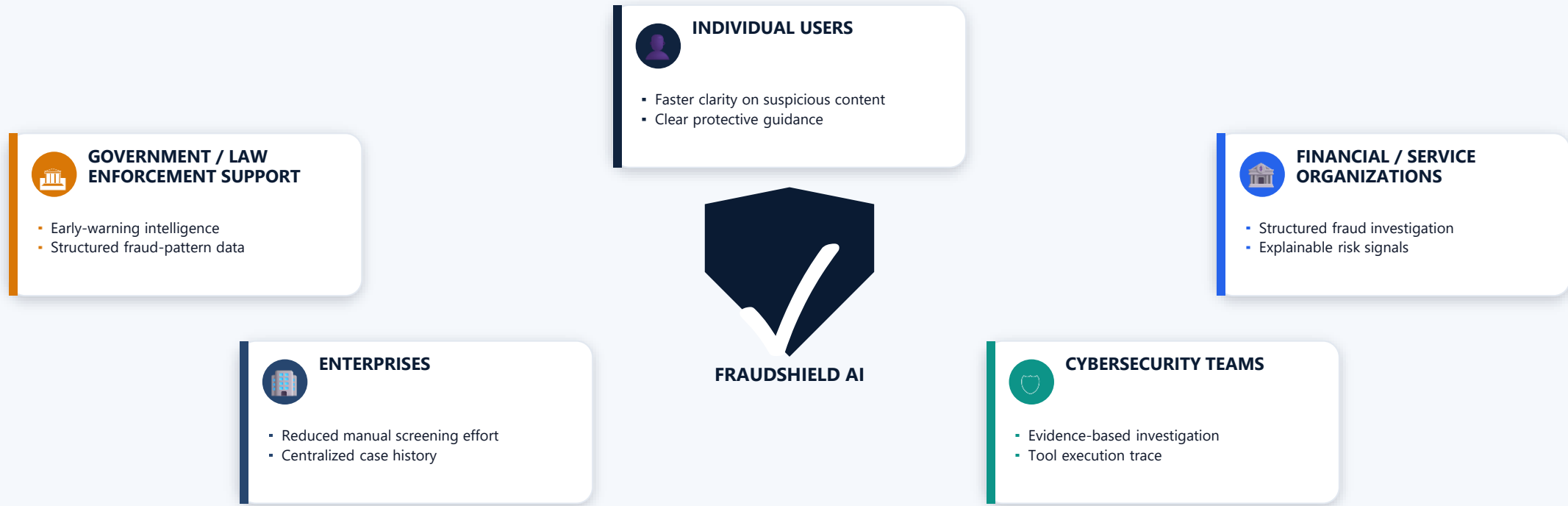
Caching + graceful fallback + lightweight local analysis

Sensitive user data

Minimal data collection + authentication + access control

PRACTICAL IMPLEMENTATION





A decision-support & early-warning platform — not a replacement for banks, police or official cybercrime systems.

IMPACT PILLARS

SOCIAL

Improves digital safety awareness and helps users respond to suspicious content.

ECONOMIC

Can reduce avoidable fraud exposure and streamline initial investigation.

SECURITY

Combines multiple signals into an explainable risk assessment.

"From suspicious input to explainable action — in one investigation."

DATASETS & TECHNICAL REFERENCES

- **UCI SMS Spam Collection**
archive.ics.uci.edu/dataset/228/sms+spam+collection
- **UCI Phishing Websites**
archive.ics.uci.edu/dataset/327/phishing+websites
- **Credit Card Fraud Detection**
Kaggle / ULB dataset
- **PhishTank**
phishtank.org
- **LangGraph Documentation**
langchain-ai.github.io/langgraph
- **Supabase Documentation**
supabase.com/docs
- **Gemini API Documentation**
ai.google.dev/gemini-api/docs

RESEARCH / SECURITY REFERENCES

- **NIST Cybersecurity Guidance**
nist.gov
- **OWASP**
owasp.org
- **CERT-In**
cert-in.org.in
- **Govt. of India Cybercrime Awareness**
cybercrime.gov.in
- **Phishing Detection Research**
Reference to be finalized
- **SMS Spam Detection Research**
Reference to be finalized
- **Fraud Detection Research**
Reference to be finalized
- **Explainable / Agentic AI Research**
Reference to be finalized

PROJECT LINKS

PROTOTYPE / DEMO
[ADD WORKING PROTOTYPE LINK]

GITHUB
[ADD GITHUB LINK]

DEMO VIDEO
[ADD VIDEO LINK]